Hertie School

Natural Language Processing

Session 6: Basic Models II
Topic models: LSA, LDA & BERTopic

GRAD-E1282 · Fall Semester 2026
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Announcements

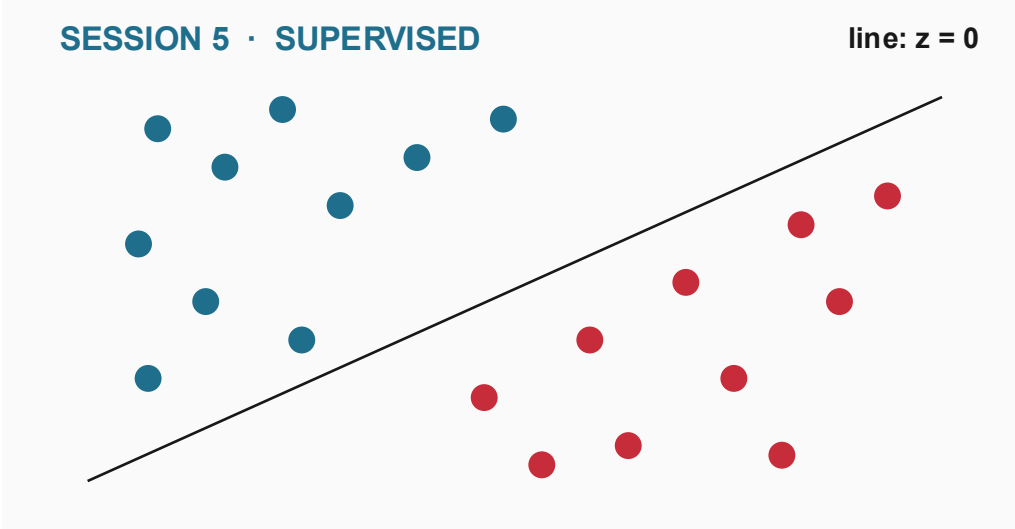
Research Note one-pager:

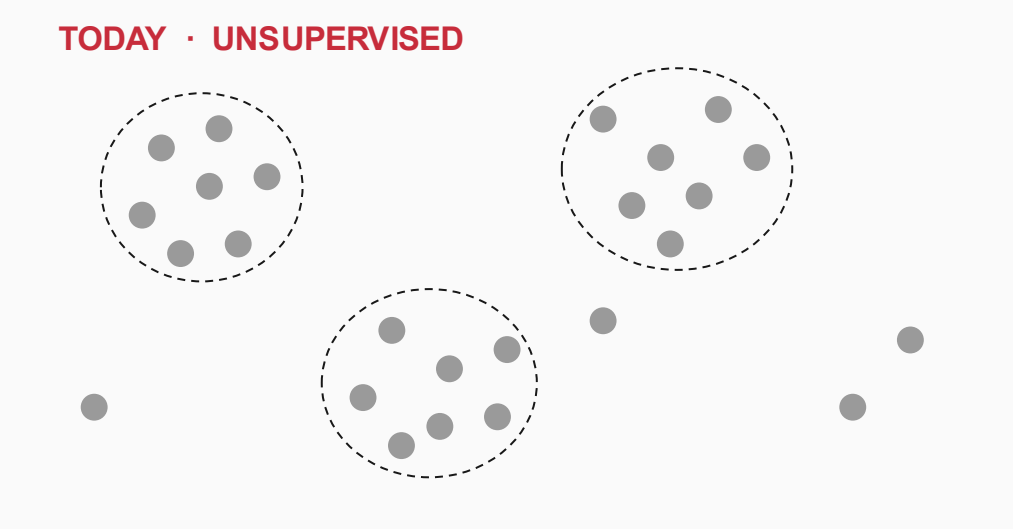
- The week after coming back from break (session 8).

Quiz 3: Week on session 9:

- Cover session 6, 7, and 8

Basic models II: when there are no labels





HOW THE LINE WAS DRAWN

$\mathbf{z = w^T x + b}$ a score: one weight per word, plus b

$\mathbf{P(y = 1 \mid x) = \sigma(z)}$ logistic regression; the line is z = 0

$\mathbf{P(c \mid x) = P(x \mid c) \cdot P(c) / P(x)}$ Bayes' theorem, Naive Bayes

WHAT WE CAN MODEL WITHOUT y

No P(y | x) to fit, because there is no y. We model how the documents themselves were written, P(x), through hidden topics:

$\mathbf{P(word \mid doc) = \Sigma_k\, P(word \mid topic\ k) \cdot P(topic\ k \mid doc)}$

Unsupervised means no answer key, not no assumptions. Every topic model decides in advance what a topic is: a direction, a word distribution, or a cluster.

Session 5 in one slide: learning with an answer key

WHAT WE ASSUMED

Labels exist, and someone paid for them. A straight boundary $w^T x + b = 0$ separates the classes well enough on sparse text.

HOW WE LEARNED IT

Minimise a loss against the labels: log-loss for logistic regression, hinge for the SVM, or count and apply Bayes' rule for Naive Bayes.

WHAT WE GOT

A weight per word and class, plus precision, recall and F1 that we could check against the truth and against a baseline.

One more look at Naive Bayes. It already described how each class writes text: P(word | class). Today we keep that story and remove the one thing it relied on, the label.

From Naive Bayes to LDA: hide the class, then mix it

One question separates the three: **who chooses the class of each word?**

NAIVE BAYES · SESSION 5

known →

class per document

per document: **one class, given by an annotator**

P(word | class) is counted from the labelled articles.

HIDE THE CLASS

hidden →

class per document

per document: **one class, inferred from the words**

Classes and P(word | class) are estimated together. BERTopic stays here: one topic per document.

MIX IT · LDA

mixed

topic per word

per document: **a mixture: every word picks a topic**

The document keeps topic proportions θ; each topic keeps word probabilities β.

The difference is where the label sits. Naive Bayes gives the whole document one label, so every word in it must come from that one class. LDA gives every word its own label, so a single article can take some words from a heat topic, some from a health topic and some from an energy-price topic.

Why topic models: four jobs

EXPLORING A CORPUS

Nobody reads 40,000 articles. Thirty word lists and ten example documents per topic are readable: a map of the corpus before any measurement.

MEASUREMENT IN SOCIAL SCIENCE

Agendas in parliamentary speeches, frames in news, themes in open survey answers. Topic proportions become variables in a regression.

CHANGE OVER TIME

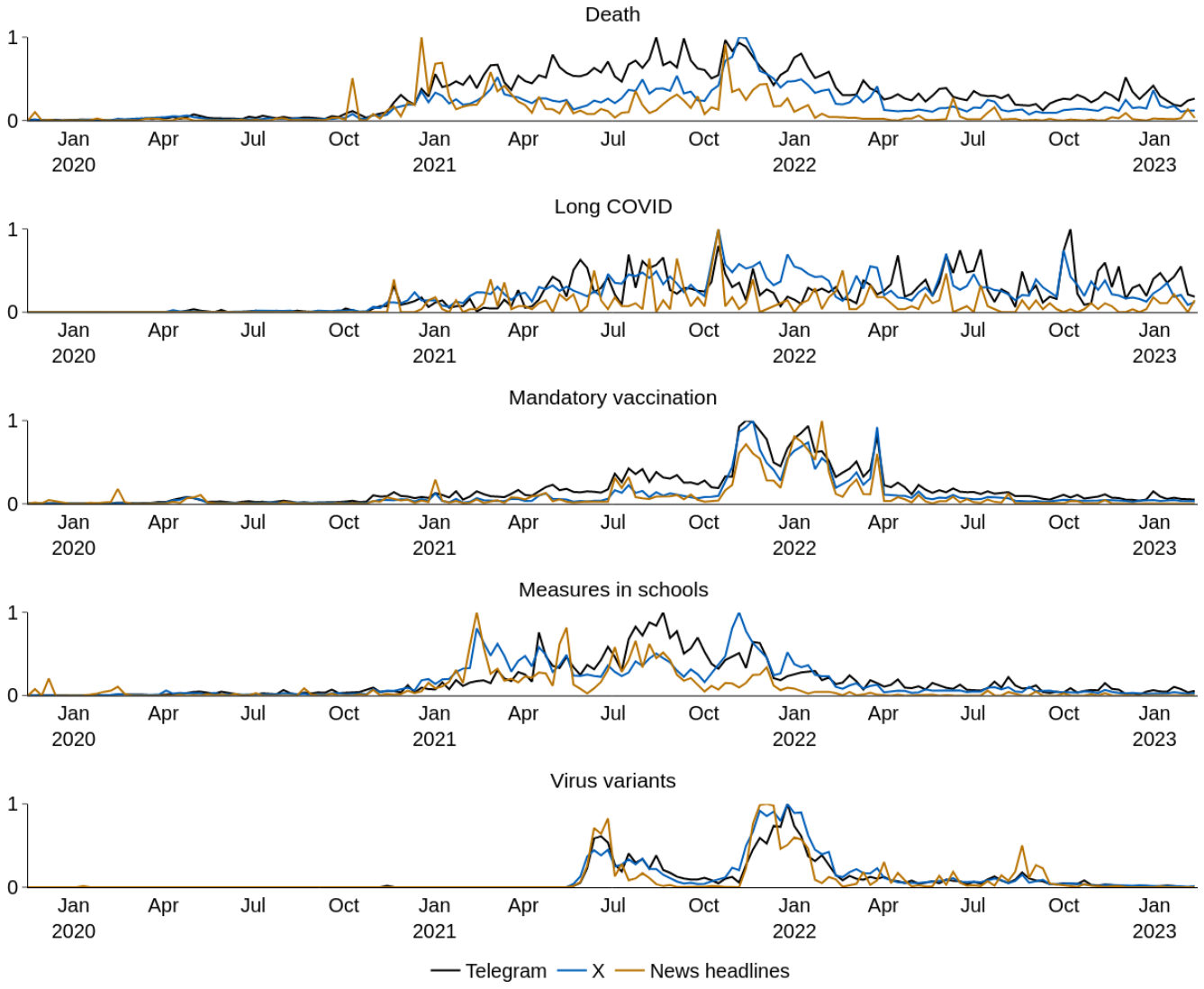
The same topic described with different words each year: how coverage of a theme grows, fades or changes its vocabulary.

FEATURES AND SEARCH

Tens of topic scores instead of thousands of word counts: compact input for last week's classifiers, and similarity search by meaning.

A topic model is an instrument for reading, not a substitute for it. Every job above ends with a person reading documents.

Examples: Topics over time of German online communities on Covid-19 vaccination discourse.



Source: Information Pathways and Voids in Critical German Online Communities During the COVID-19 Vaccination Discourse: Cross-Platform and Mixed Methods Analysis.

Authors: Silvan Wehrli, Anna-Maria Hartner, T Sonia Boender, Bert Arnrich, Christopher Irrgang

Discuss: a corpus you cannot read

1. Think of the corpus for your research note. How many documents, how long, in which languages? Could you read it all?

2. If a model gave you 20 topics, what would you want a “topic” to be: an issue, a frame, an event, a writing style?

3. Who decides that topic 7 “is” what you think it is, and what would convince a sceptical reader?

3 min in pairs → 3 min discussion.

A thought experiment: topics by hand

Six words, three topics. Each topic is a weighted sum of word scores; you choose the weights.

	cat	dog	apple	lion	NYC	love
petness	+0.3	+0.3	0	0	−0.2	+0.2
animalness	+0.1	+0.1	−0.1	+0.5	+0.1	−0.1
cityness	0	−0.1	+0.2	−0.1	+0.5	+0.1

topic vector (3 × 1) = W (3 × 6) · TF-IDF vector (6 × 1)

Read a column and you get a word vector instead: “NYC” is mostly cityness, a little animalness, and pushes against petness.

THE QUESTION THIS RAISES

Who chose +0.3 for “cat”? You did, from common sense. That does not scale to 20,000 words.

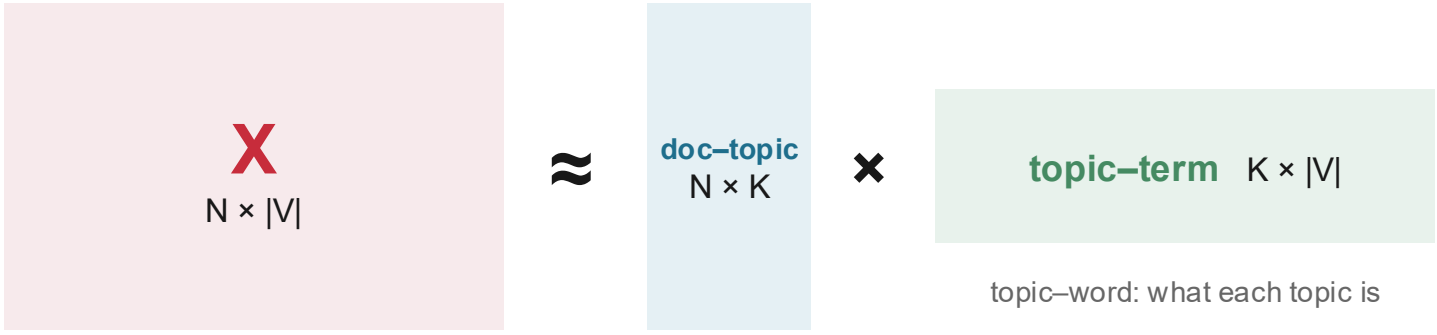
THE ANSWER SINCE SESSION 4

“You shall know a word by the company it keeps” (Firth, 1957). Words that co-occur across documents get weights on the same topic.

Every topic model learns this matrix W from co-occurrence. They differ in what they allow: negative weights (LSA), probabilities (LDA), or word lists read off clusters (BERTopic).

One matrix, two factors

Every topic model in Route A starts from Session 3's document–term matrix and splits it in two.



document–term: counts or TF-IDF (Session 3)

document–topic: how much of each topic

topic–word: what each topic is

WHY IT COMPRESSES

N = 5,000 and |V| = 19,169 give 96 million cells. With K = 16 topics the two factors hold about 390,000 numbers.

	INPUT X	THE TWO FACTORS ARE	HOW IT IS FITTED
LSA	TF-IDF	$U \cdot \Sigma$ and V^T: any sign; Σ ranks topics	truncated SVD: an exact best fit
LDA	raw counts	θ and β: probabilities, rows sum to 1	a generative story, fitted by inference

Two routes to topics: counts or embeddings

Last week the two routes fed a classifier. This week they lead to two families of topic model.

ROUTE A · COUNTS

Session 3

40,000 news articles → **document–term matrix**

BoW / TF-IDF, ~20,000 sparse columns
then: factorise into document–topic × topic–term

→ **LSA · LDA**

ROUTE B · EMBEDDINGS

Session 4

40,000 news articles → **one dense vector per document**

~384 columns, all filled
then: cluster, then read words back

→ **BERTopic**

WHAT EACH ROUTE BUYS

One model, one objective

LDA writes down how documents are generated and fits it. Mixtures come for free.

Meaning from pre-training

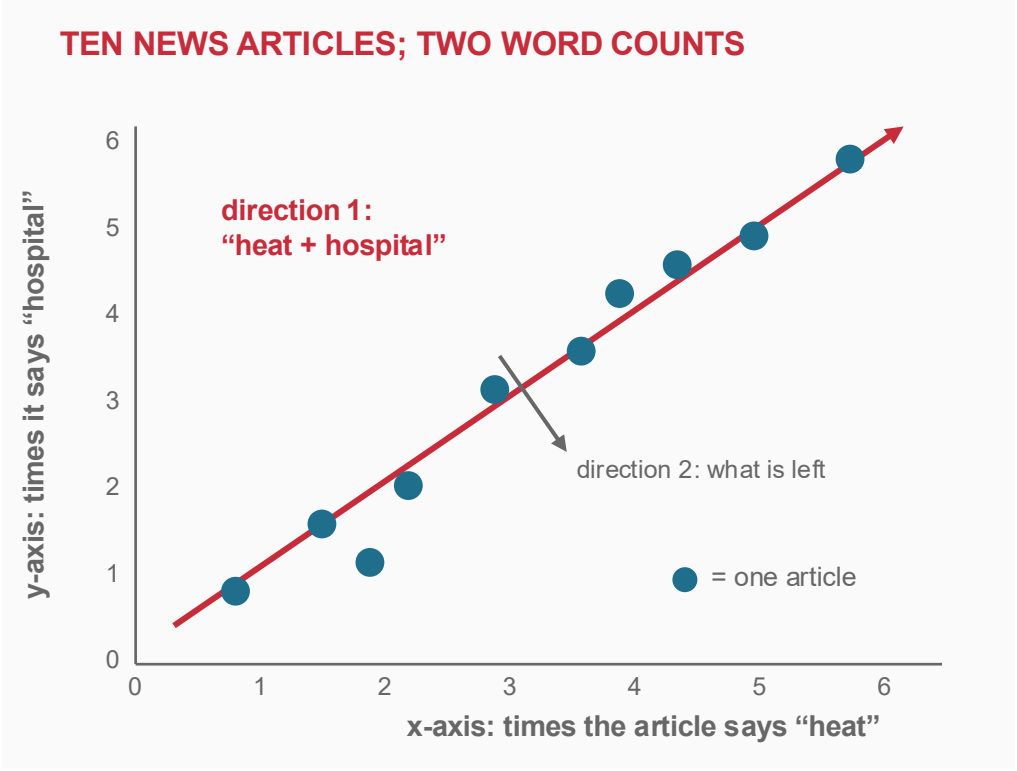
Embeddings place “heatwave” and “extreme heat” together before any topic is fitted.

The same final product

Both routes end with word lists per topic. Naming them is still your job.

Counting or embedding is a choice of representation, not of goal. After the break: one model at a time.

LSA, intuitively: words that travel together



1 · WORDS TRAVEL TOGETHER

Articles that say “heat” a lot also say “hospital” a lot. Two columns, but nearly one piece of information.

2 · ONE DIRECTION INSTEAD OF TWO WORDS

Draw the line along which the articles spread the most. Each article’s position on that line is its score for the topic.

3 · REPEAT, THEN STOP

Find the next direction at a right angle to the first, and the next. Stop after K. Each direction is one LSA topic.

You already know this idea from surveys. Fifty questions whose answers move together are summarised by two or three underlying attitudes. LSA does the same for 20,000 words, and calls the attitudes topics.

SVD in three questions: what, how much, and where

X — documents × words

	heat	hosp	pat	price	enrg	mkt
d1	3	2	2	0	0	0
d2	2	3	1	0	0	0
d3	2	1	0	1	2	0
d4	0	0	0	3	2	2
d5	0	0	0	2	3	1
d6	1	0	1	2	1	1

≈

U — documents × axes

	ax 1	ax 2
d1	+0.36	+0.60
d2	+0.31	+0.54
d3	+0.43	+0.09
d4	+0.49	−0.43
d5	+0.45	−0.38
d6	+0.39	−0.10

×

Σ — axes × axes

	ax 1	ax 2
ax 1	6.62	0
ax 2	0	5.49

×

V^T — axes × words

	heat	hosp	pat	price	enrg	mkt
ax 1	+0.44	+0.31	+0.21	+0.54	+0.54	+0.27
ax 2	+0.54	+0.53	+0.30	−0.40	−0.35	−0.25

+ 4 more axes
σ = 1.90, 1.40, 0.71, 0.23
8% of the spread **dropped**

V^T · WHAT IS EACH NEW AXIS?

A recipe of words: each row turns the six word axes into one new axis. Axis 2 = 0.54·heat + 0.53·hosp + 0.30·pat − 0.40·price − 0.35·enrg − 0.25·mkt. The rows are at right angles: the old space is only rotated.

Σ · HOW MUCH DOES EACH AXIS MATTER?

σ^2 is the spread of all documents along that axis: the sum of their squared scores. $6.62^2 = 43.8$, or 55% of the total; $5.49^2 = 30.1$, or 38%. SVD sorts the axes by σ, so "keep the first K" means "keep the strongest".

U · WHERE DOES EACH DOCUMENT SIT?

Each document's position on the new axes, scaled to length 1. Multiply by Σ to get its score: d1 = (0.36 × 6.62, 0.60 × 5.49) = (2.39, 3.27). Six word counts have become two numbers.

Rotate, then drop. Keeping all six axes reproduces X exactly: a rotation loses nothing. Correlated words (heat, hospital, patients) fold into one axis, so a few axes hold most of the spread. Dropping the weak ones costs only their σ^2, here 8%.

LSA on six articles: the two strongest directions

1 · THE COUNTS: 6 ARTICLES × 6 WORDS

	heat	hosp.	pati.	pric.	ener.	mark.
d1	3	2	2	0	0	0
d2	2	3	1	0	0	0
d3	2	1	0	1	2	0
d4	0	0	0	3	2	2
d5	0	0	0	2	3	1
d6	1	0	1	2	1	1

Blue: health words. Pink: economy words. d3 and d6 mix both.

2 · THE TWO STRONGEST DIRECTIONS

	heat	hospital	patients	price	energy	market
direction 1	+0.44	+0.31	+0.21	+0.54	+0.54	+0.27
direction 2	+0.54	+0.53	+0.30	−0.40	−0.35	−0.25

3 · READ THEM

Direction 1 is “how much of any of this”: every weight positive.

Direction 2 is a contrast: health words up, economy words down. A mixed article like d3 lands near the middle.

Two directions keep 92% of the spread in all six columns.

LSA topics are directions of contrast, not themes. That makes them good compact features and awkward to name, and it is the reason LDA insists that every weight be a probability.

Session Break

When we return: three topic models (LSA, LDA, BERTopic), each read the same four ways.

LDA, intuitively: colour every word with a highlighter

ONE ARTICLE, THREE HIGHLIGHTERS

The **heatwave** pushed **hospital admissions** up as **energy prices** hit records and **drought** hurt the **harvest**. **Doctors** urged **patients** to stay indoors.

climate health economy

COUNT THE COLOURS: THIS ARTICLE'S MIX



THE GAME LDA PLAYS ON THE WHOLE CORPUS

You have K highlighters. Colour every word in every document so that two wishes hold at once:

WISH 1 · documents use few colours

Most articles are about a few things.

WISH 2 · colours mark few distinct words

A topic should have a recognisable vocabulary.

Read the result off the colours: the share of each colour in one document is θ (its topic mix); the words each colour marked across the corpus are β (the topic).

From intuition to model: LDA's generative story

TOPIC 1 · climate

```
drought   .06
temperature   .05
emissions   .04
```

TOPIC 2 · health

```
hospital   .07
heat   .05
patients   .04
```

TOPIC 3 · economy

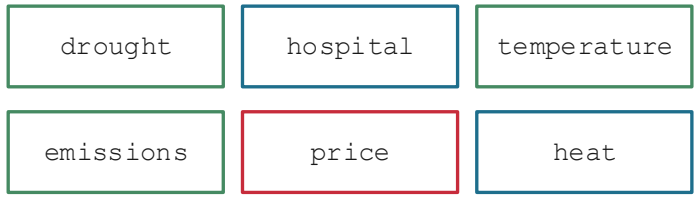
```
price   .07
energy   .05
market   .04
```

$\beta_k \sim \text{Dirichlet}(\eta)$

ONE DOCUMENT'S MIX · $\theta_d \sim \text{Dirichlet}(\alpha)$



FOR EACH WORD SLOT: TWO DICE



Roll θ_d to pick a topic z, then roll that topic's β_z to pick the word. The border shows the topic that wrote each word.

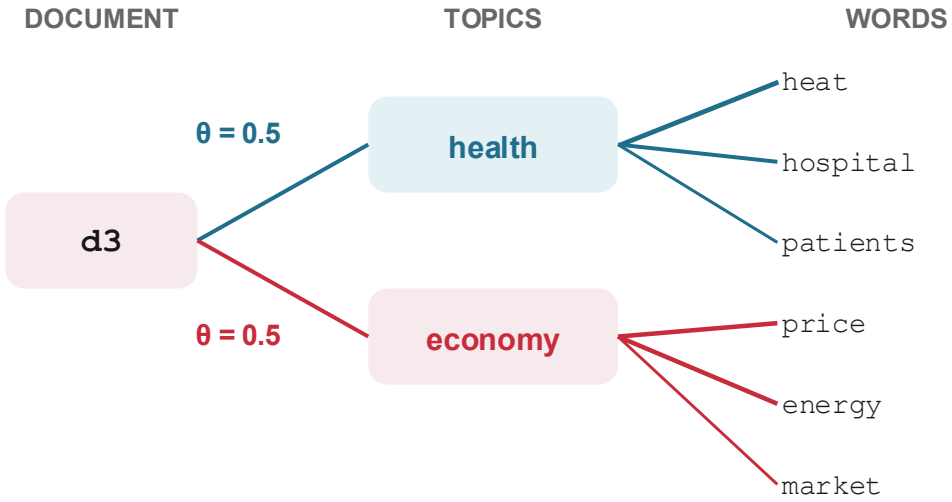
1. For each topic k, draw a word distribution β_k.
2. For each document d, draw topic proportions θ_d.
3. For each word slot, draw a topic z from θ_d, then a word from β_z.

↺ Inference runs the story backwards: given only the words, which θ and β make them most probable?

Nobody believes documents are written this way. Like the naive assumption last week, the story is false and useful: it turns "what is this corpus about?" into an estimation problem.

The equation behind the story: topics sit in the middle

$$P(\text{word } w \mid \text{document } d) = \Sigma_k \; P(w \mid \text{topic } k) \cdot P(\text{topic } k \mid d) = \Sigma_k \; \theta_{dk} \, \beta_{kw}$$



No direct line from document to word: every path passes through a topic.

line width = β

THE SAME EQUATION FOR THE WHOLE CORPUS

$$\underset{N \times |V|}{P} = \underset{N \times K}{\theta} \times \beta_{K \times |V|}$$

Rows of P are word probabilities per document. With K far smaller than the vocabulary, the topic layer is the bottleneck every document must squeeze through.

d3, from the toy LDA: `P(heat | d3) = 0.5×0.44 + 0.5×0 = 0.22`
`P(energy | d3) = 0.5×0 + 0.5×0.40 = 0.20`

Training chooses θ and β to make the observed words as probable as possible: $\text{maximise } \Pi_d \, \Pi_n \, \Sigma_k \, \theta_{dk} \, \beta_{k,w_dn}$. Same shape as LSA's $X \approx U\Sigma V^T$, but every factor is a probability, and the middle matrix is the topics.

How LDA learns: recolour one word at a time

1 Start with a random topic on every word. Then count amount of words per topic each article has (%), and how often each words carries each topic across the corpus.

2 Pick one word and remove its topic.

3 Ask two questions (counts): how much does this document already use each topic from labels of other words? How much topic k uses this word given across the corpus?

4 Reassign the word at random, in proportion to the two answers multiplied.

↺ Repeat for every word, for hundreds of sweeps. The topics settle; count them to read θ and β.

STEP 3 WITH NUMBERS · THE WORD "heat" IN ONE ARTICLE

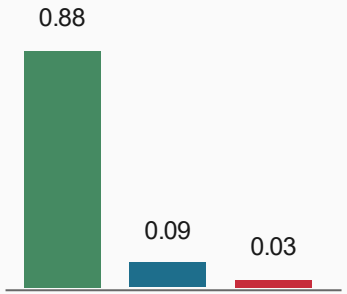
topic	document uses topic	topic uses "heat"	product	new colour
climate	4 + 0.1	0.020	0.082	**34%**
health	3 + 0.1	0.050	0.155	**65%**
economy	1 + 0.1	0.001	0.001	**<1%**

The article is mostly climate, yet "heat" will probably turn health: across the corpus, health uses "heat" far more. Both wishes vote on every word.

Two Dirichlet knobs: α for documents, η for topics

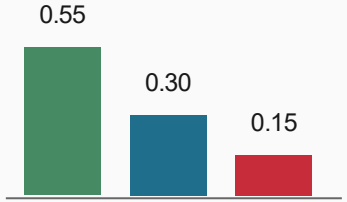
Both priors answer one question: how concentrated should a mixture be?

α = 0.1



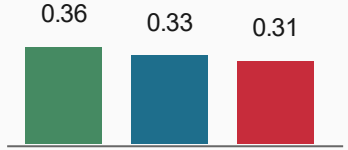
sparse: one topic dominates

α = 1



uneven mixes are common

α = 10



every document mixes everything

α · doc_topic_prior

Small α: each document uses few topics. Large α: documents blur into the average.

η · topic_word_prior

Small η: each topic concentrates on few words, sharper word lists. Large η: topics share vocabulary.

Default in scikit-learn

Both 1 / K. A sensible start; tune only after K is settled.

Priors are assumptions you can state. “News articles are mostly about one or two things” is a small α. Writing that sentence in your research note is better than leaving the default unexamined.

LDA Probability Machine

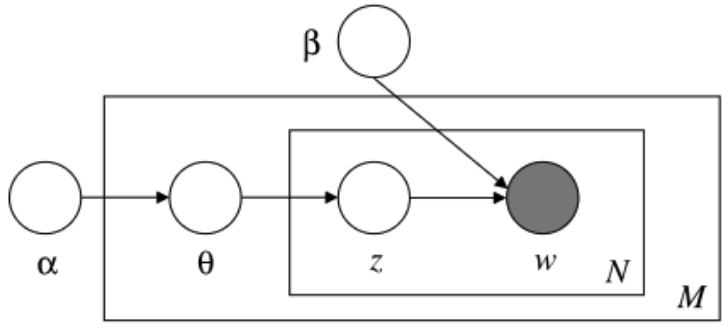


Figure 1: Graphical model representation of LDA. The boxes are "plates" representing replicates. The outer plate represents documents, while the inner plate represents the repeated choice of topics and words within a document.

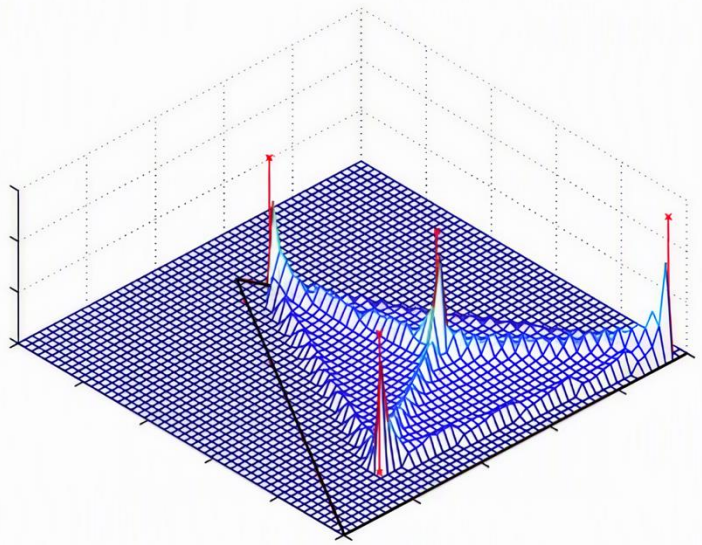


Figure 2: An example density on unigram distributions $p(w|\theta,\beta)$ under LDA for three words and four topics. The triangle embedded in the x-y plane is the 2-D simplex representing all possible multinomial distributions over three words. Each of the vertices of the trian-

Source: David M. Blei et al. "Latent Dirichlet Allocation". Published: 2003

LDA in one slide: what it assumes, optimises and outputs

WHAT IT SOLVES	Asks which K word distributions, mixed in different proportions, could have written every document in the corpus.
ASSUMPTION	Bag of words. Each document mixes K topics; each topic is a probability distribution over the vocabulary. K is fixed by you.
OBJECTIVE	Maximise the probability of the observed words, approximately: variational Bayes (scikit-learn, gensim) or Gibbs sampling.
OUTPUT	θ_d: topic proportions per document, summing to 1. β_k: word probabilities per topic.

Family: generative

Like Naive Bayes, it models how text is produced, with the class hidden.

Document → topics

A mixture of non-negative proportions: “60% climate, 30% health, 10% economy”.

Cost of the assumption

Needs enough words per document; short texts give thin mixtures. Different seeds, different topics.

Typical use: long documents that genuinely mix themes (speeches, reports, news) when you want one explicit probabilistic model of the corpus.

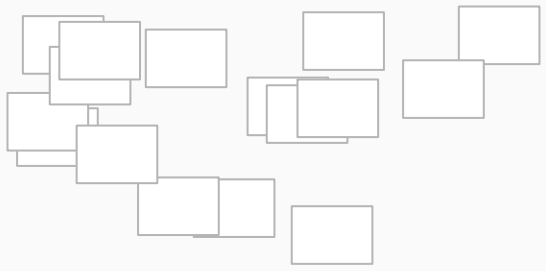
BERTopic, intuitively: sort the pile, then label each stack

1 · READ → **2 · SORT** → **3 · LABEL**

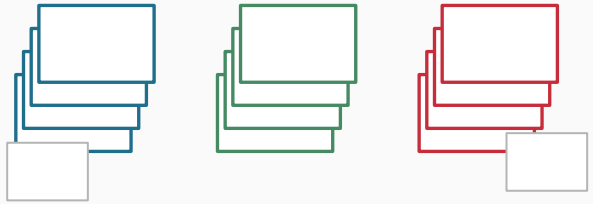
A pre-trained model gives every article a sense of what it means: one vector, Session 4 style.

Put articles with similar meaning in the same stack. Set aside the ones that fit nowhere.

Name each stack by the words that are frequent in it and rare in the other stacks.



40,000 articles, unsorted



three stacks; two set aside as −1

`hospital · patients · heat`

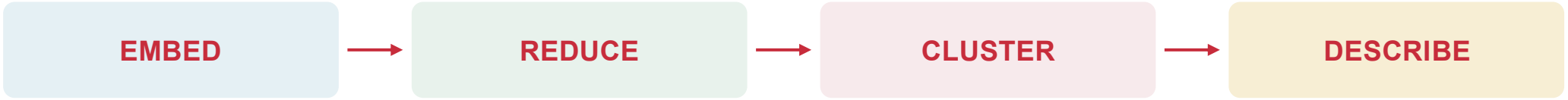
`drought · harvest · rain`

`price · energy · gas`

word lists from c-TF-IDF

LDA colours words inside documents. BERTopic sorts whole documents. And the sense of meaning comes from a model pre-trained on far more text than your corpus, not from your corpus alone.

BERTopic is a framework: four steps, four models you can swap



sentence transformer

One vector per document. Similar meaning, nearby vectors.

UMAP

From hundreds of dimensions to about 5, keeping local neighbourhoods.

HDBSCAN

Dense regions become topics; isolated documents become −1.

CountVectorizer + c-TF-IDF

Which words are frequent in this cluster and rare in the others?

WHY SEPARATE STEPS MATTER

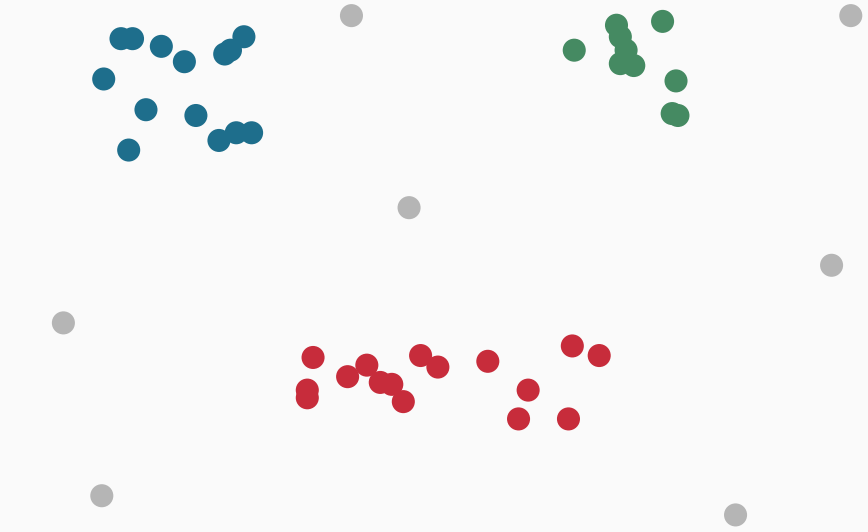
Change the word lists (stop words, bigrams) without re-embedding or re-clustering. Swap in a multilingual or domain-specific embedding model without touching the rest.

WHAT THE PAPER CHECKED

Four embedding models gave similar coherence. The small all-MiniLM-L6-v2 was about as good as the larger MPNet model, and much faster.

Every chip is a choice you report. Embedding model, UMAP and HDBSCAN settings, and the vectorizer all change the topics.

Why reduce, then cluster without K



WHY REDUCE FIRST

In hundreds of dimensions the nearest and the farthest neighbour sit at almost the same distance, so “dense” stops meaning anything. UMAP keeps local neighbourhoods in about 5 dimensions.

WHY HDBSCAN

It finds clusters of any shape and density without being told K, and it may refuse: documents that fit nowhere become −1 instead of being forced into a topic.

WHY NOT K-MEANS

K-means forces every document into a cluster and assumes round clusters around a centre. The BERTopic paper argues real topic clusters are not round.

The main knob is min_cluster_size: the smallest group you are willing to call a topic. Raise it for fewer, broader topics; lower it for more, finer ones and fewer outliers.

c-TF-IDF: Session 3's TF-IDF, with clusters as documents

SESSION 3 · ONE DOCUMENT

$$W(t, d) = tf(t, d) \cdot \log(N / df(t))$$

TODAY · ONE CLUSTER, ITS DOCUMENTS JOINED

$$W(t, c) = tf(t, c) \cdot \log(1 + A / tf(t))$$

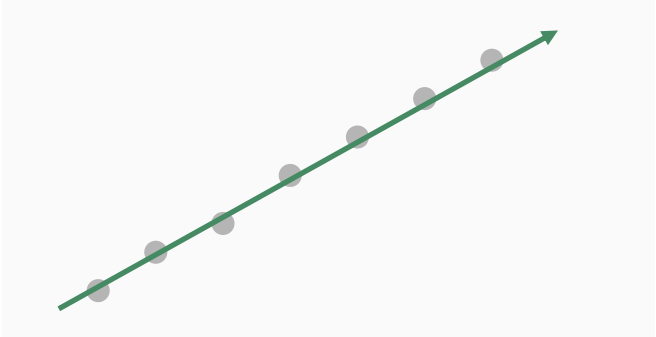
A = average number of words per cluster · tf(t) = frequency of t across all clusters · the 1 keeps every weight positive

TOY EXAMPLE: CLUSTER c_1 = "HEAT AND HEALTH", THREE CLUSTERS OF 100 WORDS EACH, SO A = 100

term	tf in c_1	tf in all clusters	log(1 + A/tf)	c-TF-IDF
heat	8	12	2.23	**17.9**
hospital	6	7	2.73	**16.4**
price	1	11	2.31	**2.3**
the	30	90	0.75	**22.4**

"the" ranks first. Common words are damped, never erased, so stop words must go in the vectorizer: CountVectorizer(stop_words="english") in the DESCRIBE step only. The embeddings still see the full text.

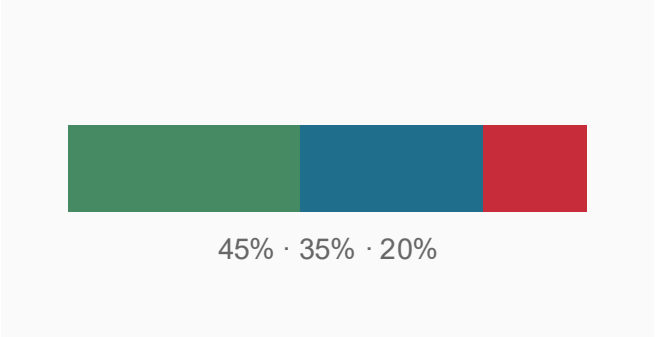
Three ideas of a topic: a direction, a recipe, a region



LSA · A DIRECTION

An axis of contrast through word space. Every document has a score on every axis, positive or negative.

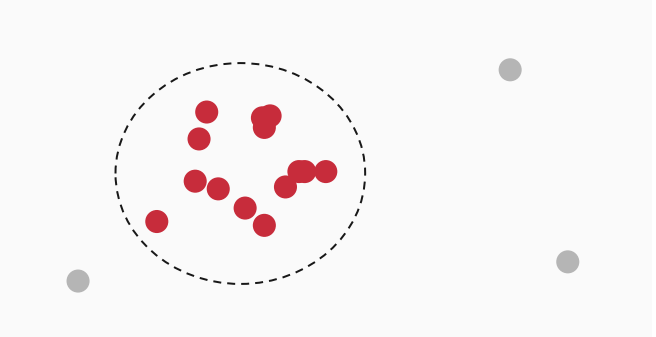
models: **the whole matrix**
meaning from: **co-occurrence in your corpus**



LDA · A RECIPE

A distribution over words. Each word in a document is drawn from one topic; a document is a mixture.

models: **every single word**
meaning from: **co-occurrence in your corpus**



BERTopic · A REGION

A dense region of documents in embedding space, described afterwards with words.

models: **whole documents**
meaning from: **pre-training on other text, then your corpus**

Two things change from left to right. The unit moves from the matrix to the word to the document, and the source of meaning moves from your corpus alone to a pre-trained language model.

Discuss: one topic or several?

1. An article reports a heatwave that closed hospitals and pushed up energy prices. How would LSA, LDA and BERTopic each represent it?

2. Take a typical document from your research corpus. Is “one topic per document” a fair assumption? If not, what would you split it into?

3. LDA reads a bag of words; BERTopic clusters embeddings of the whole text. Which would separate “heat is not a health risk” from “heat is a health risk”?

3 min in pairs → 3 min discussion.

How many topics? A choice you have to defend

LDA AND LSA · YOU SET K

Like k in k-means, K must be chosen before fitting.

Sweep a range (10, 20, … 60), fit each, and compare coherence, stability across seeds, and whether you can name the topics.

In the required reading, moving from 16 to 32 LDA topics nearly doubled the F1 of a toxicity classifier built on them.

BERTopic · K EMERGES

min_topic_size sets the smallest group that counts as a topic; the number of topics follows from the data.

To get fewer topics afterwards, nr_topics repeatedly merges the least frequent topic into its most similar one, using c-TF-IDF.

Or merge by hand after reading the topics, and say so.

`K = 5`	climate · health · economy · politics · sport
`K = 25`	heatwaves · floods · emissions policy · hospital capacity · energy prices …
`K = 200`	2022 European heatwave · Pakistan floods · one ministry's press releases …

There is no true K. Pick the granularity your research question needs, report it, and show your conclusions survive nearby values.

Evaluating without labels: which number should you report?

SESSION 5'S TOOL NEEDS A COLUMN WE DO NOT HAVE

ACTUAL = ?	PREDICTED	
	TP correctly found	FN missed
	FP false alarm	TN correctly ignored

Every cell compares a prediction with the true label. A topic model has predictions (topics) but no true labels, so the table cannot be filled.

WHAT REPLACES IT: A WORD-INTRUSION TEST

`heat · hospital · patients · **invoice** · emergency`

Show a colleague a topic's top words plus one intruder. If they spot it, the topic is coherent (Chang et al., 2009).

COHERENCE	Do a topic's top words co-occur in real documents? NPMI, from −1 to 1. The BERTopic paper's main metric.
DIVERSITY	Share of unique words across all topics' top words, 0 to 1. Low means topics repeat each other.
PERPLEXITY	How well LDA predicts held-out words. Useful for fitting, but can disagree with what people find interpretable.
STABILITY	Do the same topics reappear with another seed or a 90% subsample?
HUMAN READING	Read 10–20 documents per topic. Can a colleague spot an intruder word you inserted?

Proxies, not ground truth. The BERTopic paper says so itself: coherence and diversity only indicate quality, and NPMI may not track human judgement for neural topic models (Hoyle et al., 2021).

Discuss: when is a topic good enough?

1 Your LDA returns a topic whose top words are “said, year, also, new, people”. What went wrong, and what do you change before you rerun?

2 Coherence improves from K = 20 to K = 60, but you can only name half of the 60 topics. Which model goes into your research note, and why?

3 Two runs with different seeds give different topics. Is that a bug? What would you report to a reader?

3 min in pairs → 3 min discussion.

Where the field is going: from topics to directions

LLMs WRITE THE TOPICS

Prompt-based models generate and assign topic labels (TopicGPT, Pham et al., 2024); BERTopic can ask an LLM to name its clusters. Human review still matters (Li et al., 2025).

RESEARCHERS SEED THE TOPICS

Keyword-assisted models start from theory: keyATM (Eshima, Imai & Sasaki, 2024) or seed_topic_list in BERTopic. Discovery turns into measurement.

DIRECTIONS INSTEAD OF TOPICS

Define a concept as an axis between two poles and place every document on it (Kozlowski et al., 2019; Latent Semantic Scaling, Watanabe, 2021).

DIRECTIONS INSIDE LANGUAGE MODELS

Concepts appear as linear directions in LLM representations (Park et al., 2024); sparse autoencoders learn thousands of them without labels (Bricken et al., 2023).

Full circle back to LSA. LSA found directions and left us to guess what they meant. The new work chooses the direction first, from theory, and then has to prove it measures what it claims.

Where we are in the NLP pipeline



Data acquisition	Text extraction, normalization	Representation / features	Basic models	Neural models	Evaluation	Final evaluation + XAI	Orchestration
Not covered	Session 2	**Sessions 3–4**	SESSION 5 & 6	Sessions 7–9	Throughout	Session 10	Sessions 11-12

scikit-learn BERTopic · next: neural networks (Session 7)

Today: find structure without labels, and name it with care. Next: neural networks, which learn the features themselves.

Appendix

For reference: LDA's joint probability and inference, LSA as an SVD, and how coherence is computed.

LSA, formally: SVD and the best rank-K approximation

01	**DECOMPOSE** $\mathbf{X = U \Sigma V^T}$	U: N × r, orthonormal columns. Σ: singular values, largest first. V: |V| × r, orthonormal.
02	**TRUNCATE** $\mathbf{X_K = U_K \Sigma_K V_K^T}$	Keep the first K columns of U and V and the K largest singular values.
03	**WHY IT IS THE BEST** $\mathbf{X_K = \text{argmin over rank-K matrices of } \|X - X_K\|_F}$	Eckart–Young: no other rank-K matrix is closer in squared error. One optimum, no seeds.
04	**MAP TO SCIKIT-LEARN** $\mathbf{\text{fit_transform}(X) = U_K \Sigma_K}$, $\mathbf{\text{components_} = V_K^T}$	New documents: $x \cdot V_K$. The same projection, no refitting.
05	**PCA OR TRUNCATEDSVD?** $\mathbf{PCA = SVD \text{ of the column-centred } X}$	Centring makes a sparse TF-IDF matrix dense. TruncatedSVD skips it and stays sparse and fast.

LSA is PCA without centring, applied to TF-IDF. Topics are the directions of largest spread, which is why they are good features and hard to name.

Source: Hobson & Dyshel (2024), sections 4.3–4.4; scikit-learn user guide, TruncatedSVD.

LDA, formally: the joint probability and how it is fitted

01 THE GENERATIVE STORY AS ONE EQUATION

$$p(\theta, z, w \mid \alpha, \beta) = p(\theta \mid \alpha) \prod_n p(z_n \mid \theta)\, p(w_n \mid z_n, \beta)$$

One document with words $w_1 \ldots w_N$. $\theta \sim \mathrm{Dir}(\alpha)$; each z_n is the topic of word n.

02 WHAT WE WANT

$$p(\theta, z \mid w, \alpha, \beta) = p(\theta, z, w \mid \alpha, \beta) \,/\, p(w \mid \alpha, \beta)$$

The posterior over topic assignments. The denominator sums over all assignments: intractable.

03 VARIATIONAL BAYES

maximise a lower bound on $\log p(w \mid \alpha, \beta)$

Replace the posterior with a simpler factorised one and optimise it. scikit-learn and gensim.

04 COLLAPSED GIBBS SAMPLING

$$p(z_i = k \mid \text{rest}) \propto (n_{dk} + \alpha) \cdot (n_{kw} + \eta) \,/\, (n_k + |V|\eta)$$

Resample one word's topic at a time. Counts exclude word i itself.

05 READ THE GIBBS RULE

document likes topic × topic likes word

n_{dk}: words in d assigned to k. n_{kw}: times w is assigned to k. Rich get richer, within reason.

Only the first line is a modelling choice. Steps 02–05 are the computational price of keeping θ and z hidden.

Source: Blei, Ng & Jordan (2003); Griffiths & Steyvers (2004); scikit-learn user guide, LatentDirichletAllocation.

Coherence and diversity, formally

01 POINTWISE MUTUAL INFORMATION

$$\mathrm{PMI}(w_i, w_j) = \log\left[\, P(w_i, w_j) \,/\, \left(P(w_i)\, P(w_j) \right) \right]$$

Probabilities estimated from co-occurrence in documents or sliding windows of a reference corpus.

02 NORMALISE TO [−1, 1]

$$\mathrm{NPMI}(w_i, w_j) = \mathrm{PMI}(w_i, w_j) \,/\, \left(-\log P(w_i, w_j) \right)$$

1: always together. 0: independent. −1: never together.

03 TOPIC COHERENCE

$$C(k) = \text{mean of NPMI over pairs of the top-n words of topic } k$$

Then average over topics. The BERTopic paper uses top words via OCTIS.

04 TOPIC DIVERSITY

$$TD = \text{unique words in all top-n lists} \,/\, (K \cdot n)$$

1: no topic shares a top word with another. Dieng et al. (2020).

05 THE CAVEAT

high NPMI ≠ topics people find meaningful

Hoyle et al. (2021): automated coherence may not track human judgement for neural models.

Report coherence and diversity together, on the same reference corpus for every model you compare, and pair them with human reading.

Source: Bouma (2009); Lau et al. (2014); Dieng et al. (2020); Grootendorst (2022), section 5.3.